Assignment 1: Part D, Big Data Analysis
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Restatement and summary

Are females aged 15-45 with a certain blood type more prone to developing Polycystic Ovary Syndrome (PCOS)?

The dataset included 15 variables, both categorical and quantitative, each demonstrating varying associations with PCOS. Exploratory data analysis revealed significant associations between PCOS and factors such as age, weight, blood groups (O+ and B+), obesity, and lifestyle factors, including high fast food intake. These were confirmed by chi-square tests for categorical variables and point-biserial correlation for quantitative variables. Comparative analysis of predictive models showed that the Random Forest outperformed the Logistic Regression, achieving superior metrics across accuracy, precision, recall, and AUC. This performance is attributed to Random Forest's capability to handle complex, non-linear relationships and its better feature importance analysis. Although blood group features like B+ and O+ were prevalent among those diagnosed with PCOS, they were not as influential in predicting PCOS as they did not demonstrate a strong predictive power compared to factors such as excessive body/facial hair and skin darkening. Hence, it can be concluded that blood types are not significant predictors of PCOS.

Analysis and visualisations

Both Logistic Regression and Random Forest classifiers were refined and rigorously tested to ensure effectiveness. The Logistic Regression model was improved using Exhaustive Feature Selection (EFS), which explored all possible feature combinations to find the best set of features for higher accuracy. Both the original and refined models were evaluated with 5-fold cross-validation, assessing their performance on different subsets of the data. This process ensured the models' accuracy and their ability to generalize well to new data, preventing the omission of important features.

The Random Forest model initially using default settings with encoded categorical variables, was enhanced through hyperparameter tuning with GridSearchCV to optimize parameters, such as the number and depth of trees, to find the best configuration. The optimized model was then evaluated on a test set.

The Random Forest model outperformed Logistic Regression based on performance metrics, making it the better choice for this case. Random Forest excels in capturing complex, non-linear relationships between features, which Logistic Regression struggles with, which led to its underperformance. Additionally, Random Forest is robust against overfitting, and provides insights into which factors are most important. This leads to more accurate and consistent predictions, better resource allocation in healthcare, and informed clinical decisions, ultimately improving patient outcomes.

Table 1: Performance metrics of Logistic Regression and Random Forest models.

Model	Accuracy (%)	Precision (%)	Recall (%)	F1 Score (%)	AUC
Logistic Regression	85.05	80.45	65.56	72.24	0.8856
Random Forest	94.84	96.27	85.93	90.80	0.9907

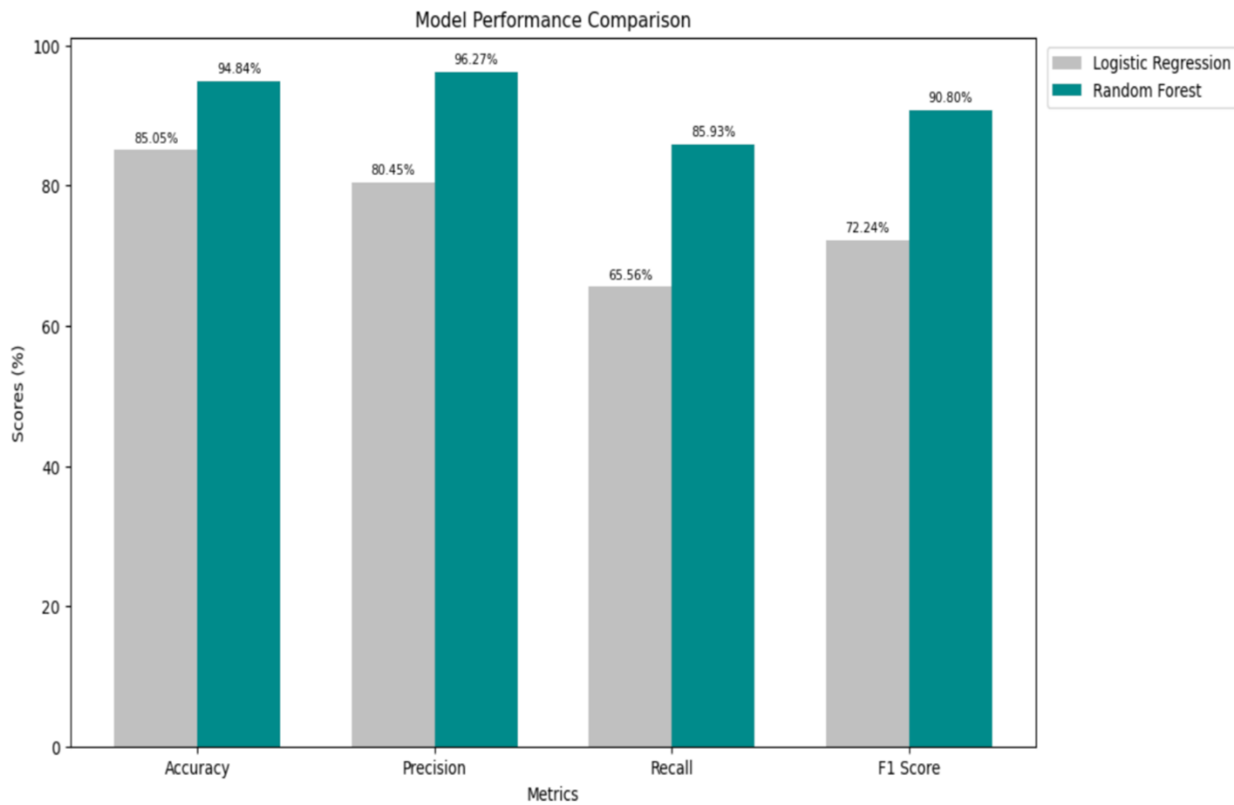

Figure 1: Grouped barchart showing the performance metrics of Logistic Regression and Random forest model.

The grouped bar chart compares the performance metrics of both models indicating Random Forest's superior overall performance.

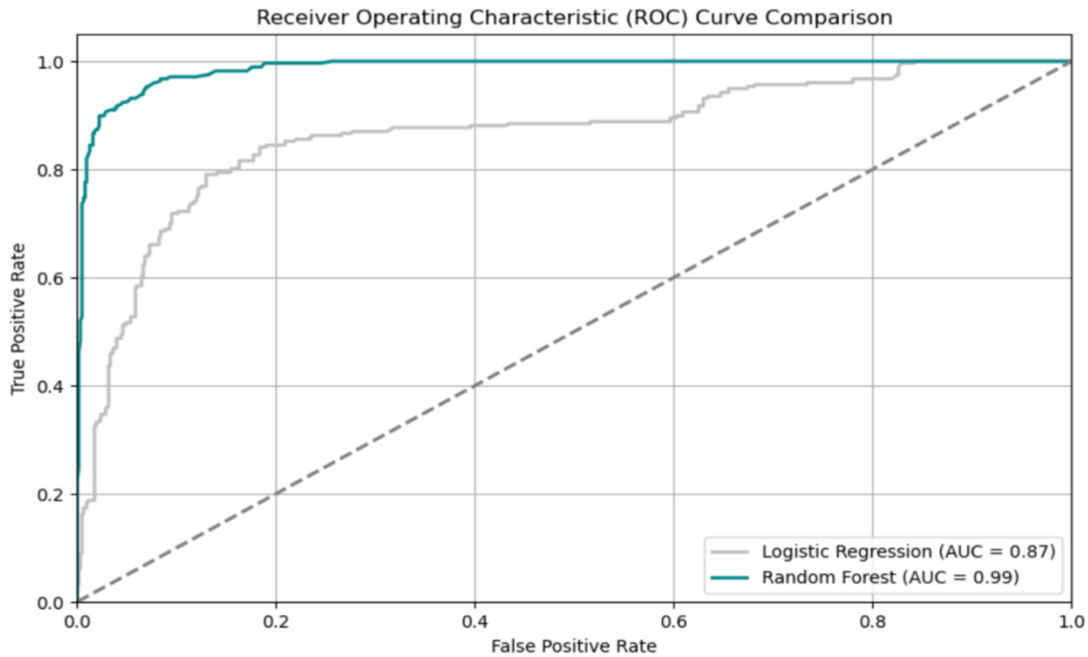

Figure 2: ROC curves of Logistic Regression and Random Forest model.

The AUC of Random Forest shows that it performed better in distinguishing whether blood type was a reason for PCOS in females compared to Logistic Regression.

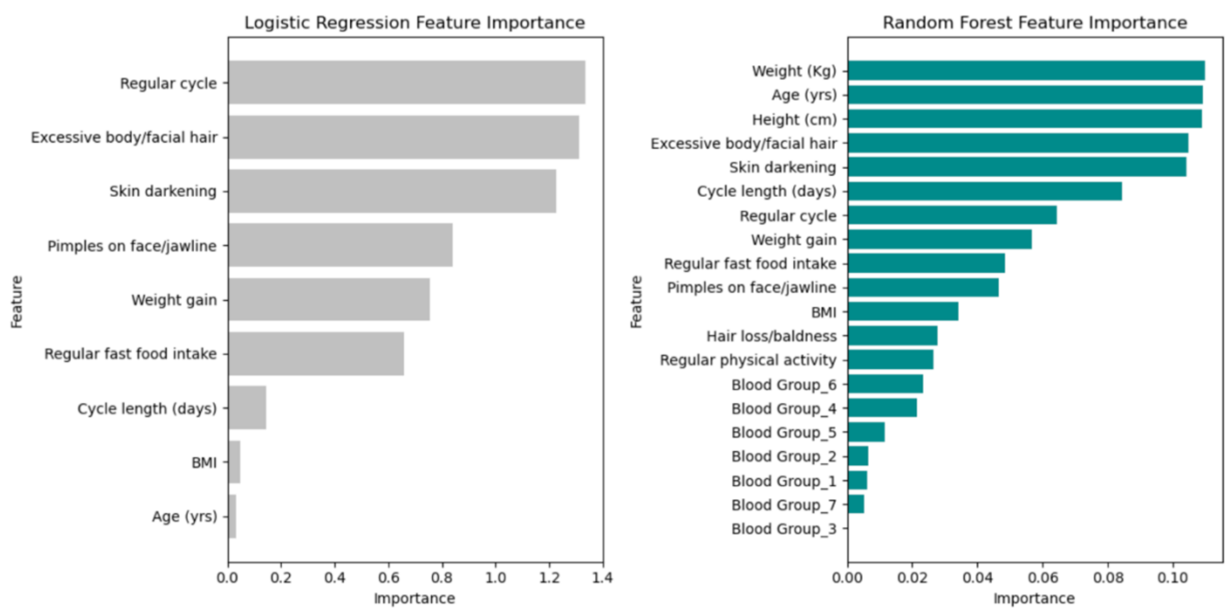

Figure 3: Feature importance plot based on Logistic Regression (selected from the best features) and Random forest model respectively.

The Logistic Regression model doesn't consider blood group to be one of the best features, similarly the Random Forest model shows least association between blood group and PCOS.

Enhancements

- Enhanced Data Collection

Gathering detailed information such as family medical history and specific health conditions can provide additional context that may affect PCOS. Key variables to include are hereditary factors like PCOS diagnoses in close family members, and other relevant health conditions such as anemia, insulin resistance, high cortisol levels, high testosterone levels, and diabetes. Collecting comprehensive generalized health data through surveys, questionnaires, or medical databases could improve model predictions.

- Feature Engineering and Selection

Creating new features, such as interaction terms between health conditions and family history, helps uncover complex patterns influencing PCOS related to blood type. Combining factors like insulin resistance and high testosterone may reveal stronger associations. Techniques like Recursive Feature Elimination (RFE) and LASSO, along with hyperparameter tuning using Python's 'scikit-learn,' can optimize models by identifying and retaining the most impactful features, thereby improving accuracy and interpretability.

Incorporating detailed family medical history and specific health conditions into the dataset and feature engineering aim to address the current models' limitations in capturing complex, non-linear relationships and subtle associations between blood type and PCOS, ultimately improving the predictive accuracy and interpretability of the models.

Conclusion and future work

The analysis highlights the Random Forest model's superior performance in predicting whether females aged 15-45 with a certain blood type are more prone to PCOS compared to Logistic Regression. Both models conclude that there is no strong association between blood type and PCOS, hence it is not accurate to say that females aged 15-45 with a certain blood type are more prone to PCOS.

Future research could explore genetic markers like insulin gene, FSHR[1], FTO[2] and biomarker interactions with blood type in PCOS, potentially revealing associations not captured by current models. Identifying strong associations with these markers may enhance model accuracy and uncover new patterns. Enhancing models with additional data and advanced methods could build on our findings, uncovering new insights. For instance, adding data on insulin resistance and employing advanced methods could further refine predictions and insights. Validating across diverse populations would ensure broader applicability and would support personalised treatments.

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[1] FSHR - follicle stimulating hormone receptor
[2] FTO – fat mass and obesity associated gene

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